

Alembic Migration Documentation for Project

1. Install Alembic

Run the following command to install Alembic:

```
pip install alembic
```

2. Initialize Alembic

initialize Alembic inside my project :

```
alembic init migration
```

This will create an `migration` folder with configuration files and a `versions` folder for migration scripts. version folder will have file that have unique revision id .i notes for each update there is a new file with unique revision id . it have two fun upgrade and downgrade that automatically updates when changes occur

Configure Alembic

1. Open `alembic.ini` and set the SQLAlchemy URL for your database:

```
sqlalchemy.url = postgresql://username:password@localhost:5432/db_name
```

2. Update `alembic/env.py` to include your models:

```
from database import Base
from config.config import DATABASE_URL

from models.user import User
from models.project import Project
from models.task import Task
```

```
from sqlalchemy import engine_from_config
from sqlalchemy import pool

from alembic import context

# this is the Alembic Config object, which provides
# access to the values within the .ini file in use.
config = context.config

config.set_main_option("sqlalchemy.url", DATABASE_URL)
target_metadata = Base.metadata
```

4. Create a Migration to Add age Column

1. Generate a new migration file:

```
alembic revision --autogenerate -m "add age column to user table"
```

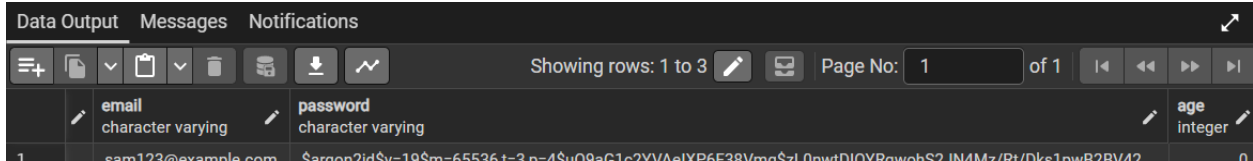
2. Alembic will create a file in alembic/versions/ folder. Verify it contains something like:

```
3. def upgrade() -> None:
4.     """Upgrade schema."""
5.     # ### commands auto generated by Alembic - please adjust! ###
6.     op.add_column('users', sa.Column('age', sa.Integer(),
7. nullable=False, server_default='20'))
8.
9. def downgrade() -> None:
10.     """Downgrade schema."""
11.     # ### commands auto generated by Alembic - please adjust! ###
12.     op.drop_column('users', 'age')
```

5. Apply the Migration

It is just like git push origin

```
alembic upgrade head
```



The screenshot shows a database table with three columns: email, password, and age. The email column is of type 'character varying' and the password column is also of type 'character varying'. The age column is of type 'integer'. The table has one row of data. The interface includes a toolbar with icons for adding, deleting, and refreshing data, and a pagination bar showing 'Showing rows: 1 to 3' and 'Page No: 1 of 1'.

	email	password	age
	character varying	character varying	integer
1	sam123@example.com	\$argon2id\$v=19\$m=65536,t=3,p=4\$u09a61e2Y3/AeLYB6F38VmpSzL0awtDIOYBawwhS2JN4Mz/BV/Dks1nwB2BV42	0

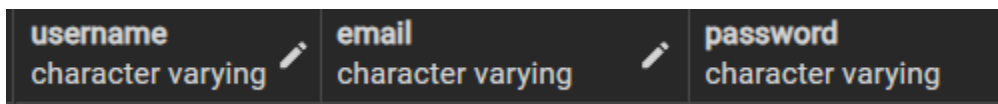
Now in pgadmin 4 we can see the User table should now have an age column.

6. Revert the Migration (Remove age Column)

To undo the last migration:

```
alembic downgrade -1
```

Check your database the age column should now be removed.



The screenshot shows a database table with three columns: username, email, and password. The username column is of type 'character varying', the email column is also of type 'character varying', and the password column is of type 'character varying'. The table has one row of data. The interface includes a toolbar with icons for adding, deleting, and refreshing data, and a pagination bar showing 'Showing rows: 1 to 3' and 'Page No: 1 of 1'.

username	email	password
character varying	character varying	character varying

Repo

https://github.com/SameenShahzad-rtc/backend_phase_1/blob/Alembic-Migration/README.md